MTH 331(A) Dr. James Ashe
Exam 2 Make-up November 19, 2012 **Name:**

1. Let $f(x) = 2e^x$.

a. Find the Taylor polynomial of order $n = 3$ for $f(x)$ centered at 0.

b. Use the Taylor polynomial from above to approximate $2e^{-0.06}$.

c. Compute the absolute error in the approximation assuming the exact value is given by a calculator.

Determine the radius of convergence of the following power series.

2. $\sum \dfrac{x^k}{0.7^k}$

Use the geometric series $\sum_{k=0}^{\infty} x^k = \dfrac{1}{1-x}$, *for* $|x| < 1$ *to find a power series representations for the following functions, and determine the interval of convergence.*

3. $f(x) = \dfrac{5}{1+x}$.

4. $f(x) = \dfrac{2x}{1-x^2}$.

Eliminate the parameter in following equations to obtain a single equation in terms of x and y.

5. $x = (t+1)^2$, $y = t+2$.

Find an equation of the line tangent to the following curve at the given point t.

6. $x = 3\sin t$, $y = 3\cos t$; $t = \pi/2$.

Give a set of parametric equations that describes the curve.

7. A bicyclist rides clockwise with a constant speed around a cricular track with radius 50 meters, completing one lap in 30 seconds.

Locate the following polar coordinates on the graph.

8. A:$\left(1.5, \dfrac{\pi}{6}\right)$, B:$\left(-1.5, \dfrac{\pi}{6}\right)$, C:$\left(1.5, \dfrac{\pi}{6} + 3\pi\right)$, D:$\left(1.5, \dfrac{\pi}{6} - 8\pi\right)$

`D:/home/jim/JamesAshe/JCSU/Calc_3/slides_Calc3/images/polar_graph_-2-2a.`

Figure 0.1: Use this graph to plot the polar coordinates.

*Express the following **polar** coordinates in **Cartesian** coordinates.*

9. $(4, 5\pi)$

*Express the following **Cartesian** coordinates in **polar** coordinates in at least **two** different ways.*

10. $\left(-1, \sqrt{3}\right)$

Multiple Choice: Identify which set of parametric curves describe the graph.

11. Parametric equations that describe the curve are

(A) $x = \cos t,\ y = \sin t,\ 0 \le t \le 2\pi.$

(B) $x = \cos t,\ y = \sin 2t,\ -\pi \le t \le \pi.$

(C) $x = \cos 2t,\ y = \sin t,\ -\pi \le t \le \pi.$

`slides_Calc3/images/cos2t_sint.png`

Find the slope of the line tangent to the following polar curves at the given point using calculus.

12. $r = 2\cos 2\theta;\ \left(2, \dfrac{\pi}{2}\right).$

Find the area of the region using calculus.

13. The region enclosed by $r = 2 + \cos 4\theta$.